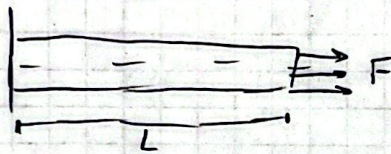


Values Chosen:

Since Cross-Sectional Area is Circular:

- Diameter (d) = 0.1382 m
- Force (F) = 300 lb_f
- Young's Modulus (E) = $8.5 \cdot 10^6$ psi
- Deflection (Δx) = 0.009 m

FBD:



Values Given:

- $\Delta L_{\max}$ (Deflection) = 0.009 in
- σ_{yield} of Aluminum = 40 ksi

Elongation Equation: $\Delta L = \frac{PL}{AE}$ (eq 1)
(P=F)

Solving For Length (L): $L = \frac{\Delta L(AE)}{P}$

$$L = \frac{(0.009 \text{ m}) \left(\frac{\pi}{4} (0.1382 \text{ m})^2 \right) (8.5 \cdot 10^6 \text{ psi})}{300 \text{ lb}_f}$$

$$\boxed{L = 3.825 \text{ in}}$$

Cross-Sectional Area (A) = $\frac{\pi}{4} (d)^2$

$$A = \frac{\pi}{4} (0.1382 \text{ m})^2 = \boxed{0.015 \text{ m}^2}$$

Stress Values:

$$\bar{\sigma}_{\text{allowable calculated}} = \frac{F}{A} = \frac{(300 \text{ lb}_f)}{0.015 \text{ in}^2} = \frac{20000 \text{ psi}}{(= 20 \text{ ksi})}$$

$$FOS_{\text{calculated}} = \frac{\bar{\sigma}_{\text{yield}}}{\bar{\sigma}_{\text{allow calculated}}} = \frac{(40 \text{ ksi})}{(20 \text{ ksi})} = \boxed{2}$$

Deflection Map:

$$\Delta L_{\text{calculated}} = 0.009 \text{ in}$$

$$\Delta L_{\text{CAD}} = 8.771 \cdot 10^{-3} \text{ in} = 0.008771 \text{ in}$$

$$\begin{aligned} \text{Percent error: } & \left| \frac{\text{Calculated} - \text{Actual}}{\text{Actual}} \right| \cdot 100 \\ &= \left| \frac{(0.008771 \text{ in}) - (0.009 \text{ in})}{(0.009 \text{ in})} \right| \cdot 100 \\ &= \boxed{2.54\%} \end{aligned}$$

Von Mises Map:

$$\bar{\sigma}_{\text{allowable calculated}} = 20 \text{ ksi}$$

$$\bar{\sigma}_{\text{allowable CAD}} = 20.42 \text{ ksi}$$

$$\begin{aligned} \text{Percent error: } & \left| \frac{\text{Calculated} - \text{Actual}}{\text{Actual}} \right| \cdot 100 \\ &= \left| \frac{(20 \text{ ksi}) - (20.42 \text{ ksi})}{(20.42 \text{ ksi})} \right| \cdot 100 \\ &= \boxed{2.06\%} \end{aligned}$$

Stress Values:

$$\sigma_{\text{allowable calculated}} = \frac{F}{A} = \frac{(300 \text{ lb})}{0.015 \text{ in}^2} = \frac{20000 \text{ psi}}{1 = 20 \text{ ksi}}$$

$$FOS_{\text{calculated}} = \frac{\sigma_{\text{yield}}}{\sigma_{\text{allowable}} = \frac{(40 \text{ ksi})}{(20 \text{ ksi})} = \boxed{2}$$

Deflection Map:

$$\Delta L_{\text{calculated}} = 0.007 \text{ in}$$

$$\Delta L_{\text{CAD}} = 8.771 \cdot 10^{-3} \text{ in} = 0.008771 \text{ in}$$

$$\begin{aligned} \text{Percent error} &= \left| \frac{\text{Calculated} - \text{Actual}}{\text{Actual}} \right| \cdot 100 \\ &= \left| \frac{(0.008771 \text{ in}) - (0.007 \text{ in})}{(0.007 \text{ in})} \right| \cdot 100 \\ &= \boxed{25.4\%} \end{aligned}$$

Von Mises Map:

$$\sigma_{\text{allowable calculated}} = 20 \text{ ksi}$$

$$\sigma_{\text{allowable CAD}} = 20.42 \text{ ksi}$$

$$\begin{aligned} \text{Percent error} &= \left| \frac{\text{Calculated} - \text{Actual}}{\text{Actual}} \right| \cdot 100 \\ &= \left| \frac{(20 \text{ ksi}) - (20.42 \text{ ksi})}{(20.42 \text{ ksi})} \right| \cdot 100 \\ &= \boxed{2.06\%} \end{aligned}$$